

WASP-36b

R-2153

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-36_140228 · created 2026-08-12T01:19:03+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2456716.71117 +/- 0.00097 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.129 +/- 0.048
Transit depth (Rp/Rs)^2	1.67 +/- 1.24 [%]
Orbital Inclination (inc)	78.18 +/- 2.85
Airmass coefficient 1 (a1)	1.04 +/- 0.61
Airmass coefficient 2 (a2)	0.0 +/- 0.7
Scatter in the residuals of the lightcurve fit is	61.64 %
Best Comparison Star	None
Optimal Aperture	3.0
Optimal Annulus	6.5
Transit Duration (day)	0.02 +/- 0.027

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.129 ± 0.048 vs 0.1368 (deviation 0.16σ)
Duration fit vs published	0.02 d vs 0.0758 d (deviation 2.07σ)
Mid-time O–C	-2.1 min
Depth SNR	2.7
Residual scatter	61.64 %

Light curve

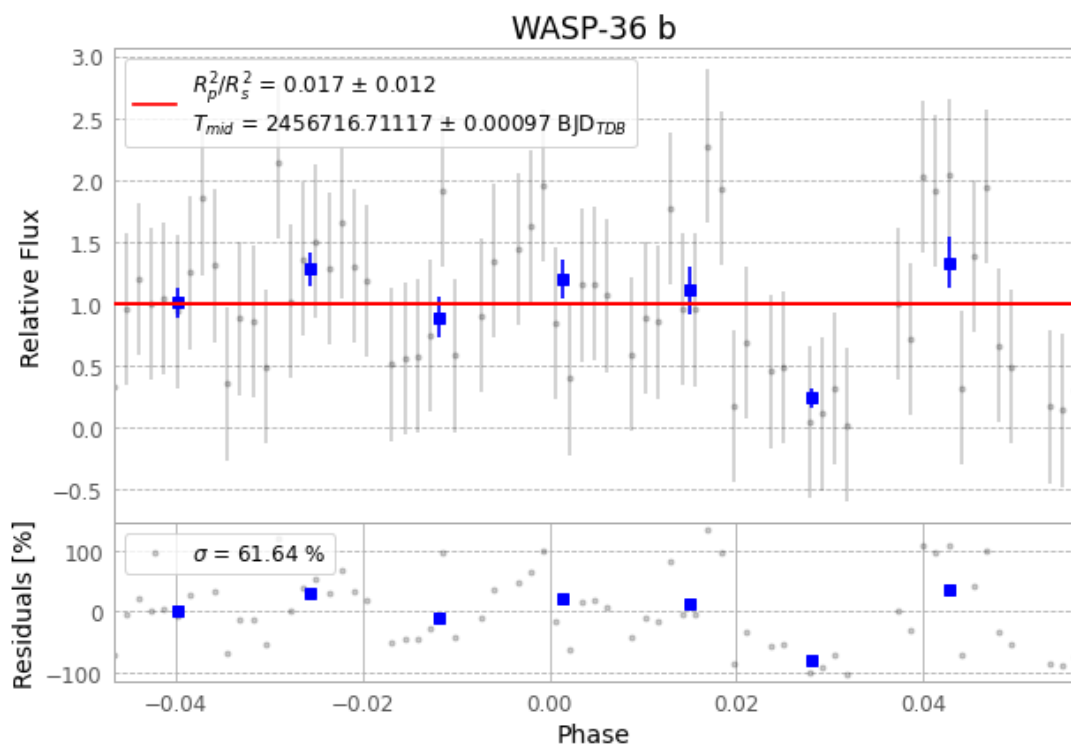


Figure 1 — FinalLightCurve_WASP-36 b_2014-02-27.png (SHA-256 2680fdb35e9676f...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [371, 392], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 e7f96f20c528d7c4...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit fd74a91

Data provenance

77 FITS frames (51 MB), WASP-36140228031212.FITS → WASP-36140228070012.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-36-140228.log (615e34c6897a...), FinalLightCurve_WASP-36 b_2014-02-27.png (2680fdb35e9...), FinalLightCurve_WASP-36 b_2014-02-27.csv (819341c37521...)

Compute resources

Wall time 182.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-12 01:19Z.